Resonant Polylogarithm Transseries: Gamma Borel Kernels, Cyclotomic Order Jets, and Sharp Dilogarithm Reversion

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Abstract

We study two operations that expose information invisible in the terminated large-argument expansion of an integer-order polylogarithm: differentiation in its order and nonlinear inversion. Starting from the classical complete Fermi–Dirac expansion, we construct an explicit lattice Borel kernel and identify the exponentially small completion with its branch signs fixed. At every positive integer order, the first order derivative has a meromorphic Borel kernel expressed by a subtracted Gamma product. Its residues recover an entire small-argument polylogarithm. A periodic-weight extension gives finite digamma and cyclotomic-polylogarithm coordinates, without discarding the zero Fourier mode. All higher order-derivative coefficients are given by harmonic Bell polynomials, and an exact Gamma quotient resolves the joint large-order and integer-resonance limit.

The main reversion result concerns $F_2(x) = -\operatorname{Li}_2(-\mathrm{e}^x)$. Its inverse about its exact quadratic core has alternating exponential sectors whose coefficients are positive rational Laurent polynomials. We prove their exact radius of convergence, identify the unique dominant square-root singularity, and obtain the explicit $N^{-3/2}$ coefficient law. On the physical curve, the sector series converges if and only if the target value is at least $\pi^2/4$; convergence at equality is absolute. The proof extends to a quadratic polylogarithm-feedback family. The article separates these derived results from classical input, supplies conventional proofs and reproducible checks, and records the limits of the source audit.

Status. Research draft with explicit proofs; not externally refereed and not Lean-formalized. The sharp reversion package and the order-jet/cyclotomic synthesis are proposed contributions, not certified claims of worldwide priority. Numerical diagnostics are not interval proofs.

Contents

1 Research setting and the main connection **3**
1.1 What is classical and what is proposed here . . . 3
1.2 A preview of the sharp threshold . . . 3
1.3 Conventions and proof dependencies . . . 4

2 The classical completion in lattice Borel coordinates **4**
2.1 An exact split of the defining integral . . . 4
2.2 A normally convergent Borel lattice . . . 5

3 Integer order derivatives and a Gamma Borel kernel **6**
3.1 The factorial tail restored by differentiation . . . 6
3.2 All higher order coefficients in harmonic coordinates . . . 8
3.3 A noncommuting resonance and large-order limit . . . 9

4 Periodic residues: finite digamma and cyclotomic coordinates **9**

5 Certified inversion of the integer-order completion **11**

6 Dilogarithm sectors are positive Laurent polynomials **12**

7 The exact radius and its unique dominant singularity **13**

8 The sharp physical threshold and an inverse identity **15**

9 A quadratic polylogarithm-feedback extension **16**

10 An audit safeguard: arithmetic averaging and inversion **17**

11 Verification, limitations, and proposed integration **18**
11.1 Exact replay and numerical diagnostics . . . 18
11.2 A numerical differentiation trap detected during the audit . . . 19
11.3 Source audit and integration boundary . . . 19

12 Further research questions and concrete next steps **20**

A Reproduction guide and result map **21**

1 Research setting and the main connection

The two repository areas meet naturally at a distinction between a formal asymptotic expansion and the analytic function that realizes it. In the normalization used here,

$$F_s(x) := -\operatorname{Li}_s(-\mathrm{e}^x), \qquad T_s(x) := F_s(-x) = -\operatorname{Li}_s(-\mathrm{e}^{-x}), \qquad x > 0. \tag{1.1}$$

The subscript is the *polylogarithm order*. The conventional normalized Fermi–Dirac index is $s-1$, not s. In particular, the inverse studied below is the inverse in x of $-\operatorname{Li}_2(-\mathrm{e}^x)$, not a choice of inverse branch in the argument of Li_2 itself.

At positive integer order m, the algebraic asymptotic expansion terminates:

$$P_m(x) = \frac{x^m}{m!} + \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{2\eta(2k)}{(m-2k)!} x^{m-2k}, \qquad \eta(z) = (1-2^{1-z})\zeta(z). \tag{1.2}$$

Nevertheless,

$$F_m(x) = P_m(x) - (-1)^m T_m(x). \tag{1.3}$$

Thus termination does not erase exponentially small information. Moreover, although the late coefficients vanish at $s = m$, their derivatives in s do not. Differentiation reveals a factorially divergent tail whose Borel poles carry the same small-argument polylogarithm. Inversion, meanwhile, converts the small tail into infinitely many coupled exponential sectors. The paper makes both mechanisms explicit.

1.1 What is classical and what is proposed here

The integral representation and functional equations for polylogarithms are classical; we use the conventions in DLMF [1]. Complete Fermi–Dirac expansions including exponentially small terms predate this article: see Garoni–Frankel–Glasser [4] and the explicit account in Gil–Segura–Temme, Section 3 and Lemma 3.1 [5]. We do *not* claim the complete Sommerfeld formula or its cosecant coefficient generator as new.

Likewise, contour Lagrange inversion, Rouché certification, and analytic square-root transfer are established methods. The repository already has a careful nonlinear Stokes-transport theorem, including countably summable perturbations, in [10]. We reuse that architecture rather than renaming it as a new general theorem. The coefficient-transfer step is classical singularity analysis [6].

The proposed additions are the assembled, model-specific statements: *the subtracted Gamma kernel for resonant order jets; its finite periodic-weight digamma dictionary; the all-order harmonic coefficient formula and detuning limit; and especially the positive Laurent formula, exact radius, unique dominant singularity, and sharp physical threshold for the dilogarithm inverse.* The latter conclusions require more than a formal Lagrange formula or a sufficient small-perturbation disk.

The current polylogarithm corpus already distinguishes accepted identities, experimental evidence, and pending analytic claims [7]. The transseries corpus also contains polylogarithmic feedback with a *linear* dominant core and nonanalytic boundary criticality [11]. Our quadratic feedback theorem is an interior-fold result with a different core. It is not presented as the first connection between polylogarithms and transseries, nor as a solution of all of that corpus's boundary problems.

1.2 A preview of the sharp threshold

Let $y > \pi^2/6$ and put

$$u = \sqrt{2y - \pi^2/3}. \tag{1.4}$$

We prove that the inverse has the sector representation

$$F_2^{-1}(y) = u + \sum_{N\geq 1} (-1)^{N-1} a_N(u) \mathrm{e}^{-Nu}, \qquad a_N(u) \in \mathbb{Q}_{>0}[u^{-1}], \tag{1.5}$$

precisely when $y \geq \pi^2/4$, including absolute convergence at equality. For $\pi^2/6 < y < \pi^2/4$, the analytic inverse still exists, but the terms of this particular sector series do not tend to zero.

This threshold is a theorem about convergence of an *exact-core transseries chart*. It is not a claim about nonexistence of the inverse, a numerical failure threshold for all algorithms, or a mere optimal truncation rule. Its proof occupies Sections 6–8.

1.3 Conventions and proof dependencies

The logarithm is real on $(0, \infty)$ and is continued with explicit lateral signs when necessary. A superscript $+$ on a Borel integral means the path passes *above* the positive Borel axis. Consequently, for $\xi > n$, the factor $1 - (\xi + \mathrm{i}0)/n$ has argument $-\pi$. This sign matters.

All asymptotic statements in the sector index N hold with the positive core parameter u fixed unless uniformity is explicitly stated. All power-series identities are identities of analytic germs in their indicated variables. The finite exact computations in the package check specified instances; the all-order assertions rest on the proofs below.

2 The classical completion in lattice Borel coordinates

2.1 An exact split of the defining integral

For $\Re s > 0$ the standard Fermi–Dirac representation is

$$F_s(x) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t^{s-1}}{1 + \mathrm{e}^{t-x}} \, \mathrm{d}t. \tag{2.1}$$

It immediately gives the following convenient exact decomposition.

Lemma 2.1 (Thermal split)**.** *For $x > 0$ and $\Re s > 0$,*

$$F_s(x) = \frac{x^s}{\Gamma(s+1)} + \frac{1}{\Gamma(s)} \int_0^\infty \frac{(x+u)^{s-1} - \mathbf{1}_{\{u<x\}}(x-u)^{s-1}}{\mathrm{e}^u + 1} \, \mathrm{d}u. \tag{2.2}$$

The identity and its order derivatives are locally uniform in the half-plane $\Re s > 0$.

Proof. Split (2.1) at $t = x$. On $t < x$ use $(1+\mathrm{e}^{t-x})^{-1} = 1 - (1+\mathrm{e}^{x-t})^{-1}$ and then set $u = x - t$. On $t > x$ set $u = t - x$. The integral of t^{s-1} from 0 to x is x^s/s. This gives the displayed formula. On a compact order set with $\Re s \geq \epsilon > 0$, endpoint singularities are bounded by a constant times $v^{\epsilon-1}(1 + |\log v|^r)$ for each fixed derivative order r; the upper endpoint has exponential decay. These bounds justify order differentiation and local uniformity. □

Expanding the odd difference near $u = 0$ yields the formal Sommerfeld series

$$\widehat{P}_s(x) = \frac{x^s}{\Gamma(s+1)} + \sum_{k\geq 1} \frac{2\eta(2k)}{\Gamma(s-2k+1)} x^{s-2k}. \tag{2.3}$$

The coefficient identity uses $\int_0^\infty u^{2k-1}/(\mathrm{e}^u + 1) \, \mathrm{d}u = \Gamma(2k)\eta(2k)$. We do not interchange an infinite, potentially divergent, expansion with (2.2); finite Taylor expansion with its remainder, or the Borel construction below, supplies the asymptotic interpretation.

2.2 A normally convergent Borel lattice

Remove the leading power by writing

$$\widehat{H}_s(x) = 1 + 2\sum_{k\geq 1} \eta(2k) s^{\underline{2k}} x^{-2k}. \tag{2.4}$$

Our Borel-minor convention is

$$\mathcal{B}\left(\sum_{n\geq 1} h_n x^{-n}\right)(\xi) = \sum_{n\geq 1} \frac{h_n}{(n-1)!}\xi^{n-1}. \tag{2.5}$$

Theorem 2.2 (Lattice kernel and exact lateral completion)**.** *The Borel minor of $\widehat{H}_s - 1$ is*

$$b_s(\xi) = s\sum_{n\geq 1} \frac{(-1)^{n-1}}{n}\left[\left(1+\frac{\xi}{n}\right)^{s-1} - \left(1-\frac{\xi}{n}\right)^{s-1}\right]. \tag{2.6}$$

This sum converges normally on compact subsets of any coherent continued sheet away from the possible singularities $\pm 1, \pm 2, \ldots$. For $\Re s > 0$, the lateral Laplace sums

$$P_s^{\pm}(x) = \frac{x^s}{\Gamma(s+1)}\left(1 + \int_0^{\infty \mathrm{e}^{\pm \mathrm{i}0}} \mathrm{e}^{-x\xi} b_s(\xi)\,\mathrm{d}\xi\right) \tag{2.7}$$

satisfy

$$P_s^{\pm}(x) = F_s(x) + \mathrm{e}^{\mp \mathrm{i}\pi s} T_s(x), \tag{2.8}$$

$$P_s^{+}(x) - P_s^{-}(x) = -2\mathrm{i}\sin(\pi s) T_s(x). \tag{2.9}$$

Thus, for the arithmetic average $P_s^{\mathrm{av}} = (P_s^{+} + P_s^{-})/2$,

$$F_s = P_s^{\mathrm{av}} - \cos(\pi s) T_s. \tag{2.10}$$

All these statements can be differentiated any fixed number of times in s, locally uniformly in $\Re s > 0$.

Proof. For $|\xi| < 1$, expand the two binomials. The coefficient of ξ^{2k-1} is

$$2s\binom{s-1}{2k-1}\sum_{n\geq 1}\frac{(-1)^{n-1}}{n^{2k}} = \frac{2\eta(2k)s^{\underline{2k}}}{(2k-1)!},$$

as required. On a fixed compact set, the bracket in the nth term is $O(n^{-1})$ as $n \to \infty$, uniformly in s on compact sets. The extra factor $1/n$ proves normal convergence. Only finitely many terms require nontrivial branch choices along any fixed compact continuation path.

On rays strictly above or below the positive axis, the kernel has at most polynomial–logarithmic growth, uniformly for s in a compact subset of $\Re s > 0$. To see this, split the lattice sum at $n \asymp |\xi|$; for larger n the preceding difference estimate applies, while the finitely many smaller terms are bounded by their power growth on the ray. The same estimates with logarithmic factors hold for order derivatives. The lateral boundary powers are locally integrable because $\Re s > 0$. They define (2.7), equivalently by ray deformation and boundary limits.

In the nth integral make the substitution $\xi = nu/x$. Absolute integrability of the large-n tail, and local integrability of the finitely many branch points, justify summing the integrals.

The geometric sum is $\sum_{n\geq 1}(-1)^{n-1}\mathrm{e}^{-nu} = (\mathrm{e}^u+1)^{-1}$. Consequently the integral in (2.7), after its prefactor is included, becomes

$$\frac{1}{\Gamma(s)}\int_0^\infty \frac{(x+u)^{s-1}-(x-u)_{\mp}^{s-1}}{\mathrm{e}^u+1}\,\mathrm{d}u.$$

For $u>x$, the last power is $\mathrm{e}^{\mp \mathrm{i}\pi(s-1)}(u-x)^{s-1}$. Comparing with (2.2) gives

$$P_s^{\pm}-F_s = -\frac{\mathrm{e}^{\mp\mathrm{i}\pi(s-1)}}{\Gamma(s)}\int_x^\infty \frac{(u-x)^{s-1}}{\mathrm{e}^u+1}\,\mathrm{d}u = \mathrm{e}^{\mp\mathrm{i}\pi s}T_s.$$

This proves the signs, the jump, and the average. The same compact-order majorants used in Lemma 2.1 justify differentiation. Finally, the Laplace integral of a finite Taylor remainder of the germ b_s gives (2.3) asymptotically; the part away from the origin is exponentially small on a fixed admissible ray. □

Corollary 2.3 (Integer resonance)**.** *For every positive integer m, b_m is a polynomial, both lateral sums are the polynomial P_m of (1.2), and*

$$F_m(x) = P_m(x) + (-1)^m \operatorname{Li}_m(-\mathrm{e}^{-x}). \tag{2.11}$$

The perturbative Stokes jump vanishes, but the physical exponential completion generally does not.

Proof. Each binomial in (2.6) is a polynomial at $s=m$. Equivalently, $1/\Gamma(m-2k+1)=0$ when $2k>m$. Equation (2.8) at $s=m$ now proves the identity. □

For example,

$$\begin{aligned}
F_1(x) &= x+\log(1+\mathrm{e}^{-x}),\\
F_2(x) &= \frac{x^2}{2}+\frac{\pi^2}{6}+\operatorname{Li}_2(-\mathrm{e}^{-x}),\\
F_3(x) &= \frac{x^3}{6}+\frac{\pi^2 x}{6}-\operatorname{Li}_3(-\mathrm{e}^{-x}).
\end{aligned}$$

These checks fix the parity convention independently of Borel terminology.

3 Integer order derivatives and a Gamma Borel kernel

3.1 The factorial tail restored by differentiation

Write

$$A_k(s) = \frac{2\eta(2k)}{\Gamma(s-2k+1)}. \tag{3.1}$$

If $n=2k-m\geq 1$, Gamma reflection gives the exact local identity

$$A_k(m+\epsilon) = 2(-1)^{m+1}\eta(2k)\frac{\sin(\pi\epsilon)}{\pi}\Gamma(n-\epsilon). \tag{3.2}$$

In particular,

$$A_k(m)=0, \qquad A_k'(m) = 2(-1)^{m+1}\eta(2k)\Gamma(2k-m). \tag{3.3}$$

Thus the first order derivative has infinitely many nonzero late coefficients even though the undifferentiated series terminates.

Define

$$K_m(x) = \frac{x^m}{m!}(\log x - \psi(m+1)) + \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{2\eta(2k)x^{m-2k}}{(m-2k)!}(\log x - \psi(m-2k+1)). \tag{3.4}$$

This is the finite power–log part of the first order derivative. Put $\tau_0 = 1$ and $\tau_{2k} = 2\eta(2k)$ for $k \geq 1$.

Theorem 3.1 (Subtracted Gamma kernel)**.** *For $m \geq 1$ the ordinary Borel minor of the late part of $\partial_s \widehat{P}_s|_{s=m}$ is*

$$B_m(\xi) = (-1)^{m+1}\xi^{-m-1}\left\{\Gamma(1+\xi)\Gamma(1-\xi) - \sum_{k=0}^{\lfloor m/2 \rfloor} \tau_{2k}\xi^{2k}\right\} \tag{3.5}$$

$$= (-1)^{m+1}\xi^{-m-1}\left\{\frac{\pi\xi}{\sin \pi\xi} - \sum_{k=0}^{\lfloor m/2 \rfloor} \tau_{2k}\xi^{2k}\right\}. \tag{3.6}$$

The apparent singularity at zero is removable. Its positive poles have residues

$$\operatorname{Res}_{\xi=n} B_m(\xi) = \frac{(-1)^{m+n+1}}{n^m}, \qquad n \geq 1. \tag{3.7}$$

For $J_m(x) = \partial_s F_s(x)|_{s=m}$, one has the exact identity

$$J_m(x) = K_m(x) + \mathrm{PV}\int_0^\infty \mathrm{e}^{-x\xi} B_m(\xi)\,\mathrm{d}\xi - (-1)^m\, \partial_s T_s(x)|_{s=m}\,. \tag{3.8}$$

Here the principal value is the arithmetic average of the two lateral integrals, equivalently the sum of symmetric local principal values at the positive integer poles. In particular,

$$\int_+ \mathrm{e}^{-x\xi} B_m(\xi)\,\mathrm{d}\xi - \int_- \mathrm{e}^{-x\xi} B_m(\xi)\,\mathrm{d}\xi = -2\pi\mathrm{i}(-1)^m T_m(x). \tag{3.9}$$

Proof. The tail in (3.3) has Borel minor

$$2(-1)^{m+1} \sum_{k>m/2} \eta(2k)\xi^{2k-m-1}. \tag{3.10}$$

The classical generating identity $\pi\xi/\sin\pi\xi = 1 + 2\sum_{k\geq 1}\eta(2k)\xi^{2k}$ and Gamma reflection [2] give (3.5)–(3.6). The first exponent in (3.10) is nonnegative. The residue follows from $(\sin\pi\xi)'|_{\xi=n} = \pi(-1)^n$.

For the exact, rather than only formal, assertion, let $c_m = \partial_s b_s|_{s=m}$. Differentiating the lattice kernel m times in ξ gives

$$\frac{c_m^{(m)}(\xi)}{m!} = \sum_{n\geq 1} \frac{(-1)^{n-1}}{n^m}\left(\frac{1}{n+\xi} - \frac{(-1)^m}{n-\xi}\right) = B_m(\xi). \tag{3.11}$$

Indeed, the mth derivative of $(1+a\xi)^{m-1}\log(1+a\xi)$ is $(m-1)!a^m/(1+a\xi)$; polynomial terms disappear. The series converges normally away from its poles.

On either tilted Laplace ray, repeated integration by parts yields

$$x^m \int \mathrm{e}^{-x\xi} c_m(\xi)\,\mathrm{d}\xi = \int \mathrm{e}^{-x\xi} c_m^{(m)}(\xi)\,\mathrm{d}\xi + \sum_{j=0}^{m-1} x^{m-1-j} c_m^{(j)}(0).$$

The polynomial endpoint contributions, combined with the derivative of $x^s/\Gamma(s+1)$ in (2.7), are exactly K_m. For verification, the endpoint at $j = 2k-1 < m$ contributes

$$\frac{2\eta(2k)x^{m-2k}}{(m-2k)!}(\psi(m+1)-\psi(m-2k+1)),$$

which changes the common $\log x - \psi(m+1)$ to the term in (3.4). Therefore

$$\partial_s P_s^{\pm}(x)\big|_{s=m} = K_m(x) + \int_{\pm} \mathrm{e}^{-x\xi} B_m(\xi)\,\mathrm{d}\xi. \tag{3.12}$$

Taking the order derivative of (2.8), and then averaging, proves (3.8). Subtraction proves (3.9). The sums of pole contributions are absolutely convergent after multiplication by e^{-nx}, so the indicated global principal value is well defined. □

The first two kernels illustrate the different local parity:

$$B_1(\xi) = \frac{\pi}{\xi\sin\pi\xi} - \frac{1}{\xi^2}, \tag{3.13}$$

$$B_2(\xi) = -\frac{1}{\xi^3}\left(\frac{\pi\xi}{\sin\pi\xi} - 1 - \frac{\pi^2\xi^2}{6}\right). \tag{3.14}$$

For the upper lateral integral,

$$J_m = K_m + \int_{+} \mathrm{e}^{-x\xi} B_m(\xi)\,\mathrm{d}\xi - (-1)^m(T'_m - \mathrm{i}\pi T_m), \tag{3.15}$$

where T'_m denotes an *order derivative*, not an x derivative. Equation (3.15) is the form used in the independent complex-ray numerical checks.

3.2 All higher order coefficients in harmonic coordinates

The complete exponential Bell polynomials are specified by

$$\exp\left(\sum_{j\ge1} b_j\frac{z^j}{j!}\right) = \sum_{r\ge0}\mathcal{Y}_r(b_1,\dots,b_r)\frac{z^r}{r!}, \qquad \mathcal{Y}_0 = 1. \tag{3.16}$$

Let $H_n^{(j)} = \sum_{\ell=1}^{n}\ell^{-j}$ and $H_n = H_n^{(1)}$, with $H_0^{(j)} = 0$.

Theorem 3.2 (Harmonic Bell formula for every resonant jet)**.** *Fix $m \ge 1$, $k > m/2$, $n = 2k-m$, $r \ge 1$, and $x > 0$. Define*

$$b_1 = \log x + \gamma - H_{n-1}, \tag{3.17}$$

$$b_j = (j-1)!\left((-1)^{j+1}\zeta(j) - H_{n-1}^{(j)}\right), \qquad j \ge 2. \tag{3.18}$$

Then the individual late coefficient has the exact order-derivative formula

$$\partial_s^r\left(A_k(s)x^{s-2k}\right)\Big|_{s=m} = 2(-1)^{m+1}\eta(2k)\Gamma(n)x^{-n}r\mathcal{Y}_{r-1}(b_1,\dots,b_{r-1}). \tag{3.19}$$

The Stokes jump of the full rth order jet is obtained by differentiating $-2\mathrm{i}\sin(\pi s)T_s$ exactly r times at $s = m$.

Proof. Divide (3.2), after multiplication by $x^{-n+\epsilon}$, by $2(-1)^{m+1}\eta(2k)\Gamma(n)x^{-n}$. The result is

$$\epsilon\,\frac{\sin\pi\epsilon}{\pi\epsilon}\,\frac{\Gamma(n-\epsilon)}{\Gamma(n)}x^{\epsilon}.$$

The logarithm of the factor multiplying ϵ is analytic near zero. Its first derivative is $\log x - \psi(n) = b_1$. For $j \geq 2$, use

$$\psi^{(j-1)}(n) = (-1)^j(j-1)!(\zeta(j) - H^{(j)}_{n-1})$$

and $\log(\sin\pi\epsilon/(\pi\epsilon)) = -\sum_{\ell\geq 1}\zeta(2\ell)\epsilon^{2\ell}/\ell$. The jth derivative is precisely (3.18). The Bell generating identity now gives (3.19), since $r!/(r-1)! = r$. The full jump statement follows from the locally uniform order differentiation already proved in Theorem 2.2. □

For $r = 1, 2, 3$ the respective Bell factors are 1, $2b_1$, and $3(b_1^2 + b_2)$. The sign $b_2 = -\zeta(2) - H^{(2)}_{n-1}$ is worth noting: it includes the contribution of the sine factor, not only the Gamma derivative.

3.3 A noncommuting resonance and large-order limit

Corollary 3.3 (Uniform coefficient detuning)**.** *With $n = 2k - m \to \infty$, and the value at $\epsilon = 0$ understood by continuity,*

$$\frac{A_k(m+\epsilon)}{\epsilon A_k'(m)} = \frac{\sin\pi\epsilon}{\pi\epsilon}\,\frac{\Gamma(n-\epsilon)}{\Gamma(n)} = \frac{\sin\pi\epsilon}{\pi\epsilon}\,n^{-\epsilon}(1 + O(n^{-1})). \tag{3.20}$$

The last estimate is uniform for ϵ in any fixed compact subset of $\mathbb{C}$. If $\epsilon \to 0$ and $\epsilon\log n \to \lambda \in \mathbb{C}$, then

$$\frac{A_k(m+\epsilon)}{\epsilon A_k'(m)} \longrightarrow \mathrm{e}^{-\lambda}. \tag{3.21}$$

For the coefficient including x^{s-2k}, the right side of (3.20) is multiplied by x^{ϵ}.

Proof. The first identity is (3.2). Uniform Stirling expansion of the Gamma quotient [3], with n tending to infinity on the positive real axis and bounded ϵ, gives the second. The limit follows from $n^{-\epsilon} = \exp(-\epsilon\log n)$ and the removable value 1 of the sinc factor. This is a coefficient-level uniform statement; it does not by itself assert uniform optimal truncation of the full function. □

4 Periodic residues: finite digamma and cyclotomic coordinates

The lattice kernel admits an arithmetic extension that directly meets the finite-Fourier polylogarithm–polygamma bridge of the repository [8]. The finite Fourier transform itself is classical; the point here is to apply it to a *meromorphic Borel kernel* and its entire exponentially small jump, including the zero Fourier coefficient.

Let $c = (c_n)_{n\geq 1}$ have period $d \geq 1$, with complex entries. Set

$$L_c(j) = \sum_{n\geq 1}\frac{c_n}{n^j} = d^{-j}\sum_{a=1}^{d} c_a\zeta(j, a/d), \qquad j \geq 2, \tag{4.1}$$

$$T_{m,c}(x) = \sum_{n\geq 1}\frac{c_n\mathrm{e}^{-nx}}{n^m}, \qquad x > 0. \tag{4.2}$$

No condition $\sum_a c_a = 0$ is imposed.

Theorem 4.1 (Periodic Borel-kernel dictionary)**.** *For $m \geq 1$, the normally convergent meromorphic sum*

$$B_{m,c}(\xi) = \sum_{n\geq 1} \frac{c_n}{n^m}\left(\frac{1}{n+\xi} - \frac{(-1)^m}{n-\xi}\right) \tag{4.3}$$

has the finite digamma representation

$$B_{m,c}(\xi) = \frac{(-1)^m}{\xi^m}\Bigg\{\frac{1}{d}\sum_{a=1}^{d} c_a\left[\psi\Big(\frac{a-\xi}{d}\Big) - \psi\Big(\frac{a+\xi}{d}\Big)\right] + 2\sum_{k=1}^{\lfloor m/2\rfloor} L_c(2k)\xi^{2k-1}\Bigg\}. \tag{4.4}$$

The singularity at zero is removable. The positive-pole residues and the lateral Laplace jump are

$$\operatorname{Res}_{\xi=n} B_{m,c}(\xi) = (-1)^m \frac{c_n}{n^m}, \tag{4.5}$$

$$\int_{+} \mathrm{e}^{-x\xi} B_{m,c}(\xi)\,\mathrm{d}\xi - \int_{-} \mathrm{e}^{-x\xi} B_{m,c}(\xi)\,\mathrm{d}\xi = -2\pi\mathrm{i}(-1)^m T_{m,c}(x). \tag{4.6}$$

Writing $\omega = \mathrm{e}^{2\pi\mathrm{i}/d}$ and $\widehat{c}_p = d^{-1}\sum_{a=1}^{d} c_a\omega^{-pa}$ gives the finite identity

$$T_{m,c}(x) = \sum_{p=0}^{d-1} \widehat{c}_p\,\mathrm{Li}_m(\omega^p \mathrm{e}^{-x}). \tag{4.7}$$

The constants in (4.4) also have finite polygamma coordinates:

$$L_c(2k) = \frac{1}{d^{2k}(2k-1)!}\sum_{a=1}^{d} c_a\psi^{(2k-1)}(a/d). \tag{4.8}$$

Proof. Introduce the absolutely convergent, subtracted resolvent

$$S_c(z) = \sum_{n\geq 1} c_n\left(\frac{1}{n+z} - \frac{1}{n}\right) = \frac{1}{d}\sum_{a=1}^{d} c_a\left[\psi(a/d) - \psi((a+z)/d)\right]. \tag{4.9}$$

The equality follows by grouping the standard digamma difference series in residue classes. Its convergence does not require a zero mean. For $R_m(z) = \sum_{n\geq 1} c_n/[n^m(n+z)]$ one has $R_1 = -S_c/z$ and, for $m \geq 2$, $R_m = (L_c(m) - R_{m-1})/z$. Hence

$$R_m(z) = \sum_{j=2}^{m} \frac{(-1)^{m-j}L_c(j)}{z^{m-j+1}} + \frac{(-1)^m S_c(z)}{z^m}.$$

Now form $R_m(\xi) - (-1)^m R_m(-\xi)$. The odd-$j$ terms cancel and (4.4) follows from (4.9). The local Taylor expansion of (4.3) is

$$B_{m,c}(\xi) = 2(-1)^{m+1}\sum_{k>m/2} L_c(2k)\xi^{2k-m-1}, \qquad |\xi| < 1, \tag{4.10}$$

which proves removability. This is also the resonant order-jet kernel of (2.6) with $(-1)^{n-1}$ replaced by c_n.

The residue in (4.5) is immediate from its single pole term. On tilted rays the finite digamma expression has at most polynomial–logarithmic growth. Closing the upper-minus-lower contour in the Laplace-decaying sector gives minus $2\pi\mathrm{i}$ times the positive residue sum. One can take outer radii avoiding the integer poles; the exponential factor makes the outer arcs and residue tails tend to zero. This proves (4.6). Finally, $c_n = \sum_p \widehat{c}_p\omega^{pn}$ and absolute convergence of (4.2) give (4.7). Grouping the defining Dirichlet series gives (4.1); the standard Hurwitz-zeta/polygamma identity gives (4.8). □

Remark 4.2 (A nonzero-mean example). Take $c_n = 1$ for $n \equiv 1 \pmod 3$ and zero otherwise. Then

$$B_{1,c}(\xi) = -\frac{1}{3\xi}\left[\psi\Big(\frac{1-\xi}{3}\Big) - \psi\Big(\frac{1+\xi}{3}\Big)\right],$$

and

$$T_{m,c}(x) = \frac{1}{3}\sum_{p=0}^{2} \omega^{-p}\,\mathrm{Li}_m(\omega^p \mathrm{e}^{-x}).$$

The $p = 0$ term is $\mathrm{Li}_m(\mathrm{e}^{-x})/3$ and cannot be dropped. For the alternating sequence $c_n = (-1)^{n-1}$, the theorem reduces to Theorem 3.1. Thus one and the same residue dictionary has Gamma, polygamma, Hurwitz-zeta, and cyclotomic-polylogarithm forms.

5 Certified inversion of the integer-order completion

At integer order, a finite algebraic core is known exactly. Retaining it unexpanded avoids the artificial proliferation of algebraic terms that would result from first approximating its inverse. Fix $m \geq 1$ and $u > 0$, put $y = P_m(u)$, and regard q as an independent complex parameter. The inverse displacement is the germ $\delta(q)$ with $\delta(0) = 0$ satisfying

$$P_m(u+\delta) - P_m(u) + (-1)^m\,\mathrm{Li}_m(-q\mathrm{e}^{-\delta}) = 0. \tag{5.1}$$

The physical substitution is $q = \mathrm{e}^{-u}$. Let

$$Q_{m,u}(z) = \frac{P_m(u+z) - P_m(u)}{z}, \qquad Q_{m,u}(0) = P_m'(u) > 0, \tag{5.2}$$

and define the finite composition sum

$$C_{N,k}^{(m)} = [t^N]\,\mathrm{Li}_m(t)^k = \sum_{\substack{n_1+\cdots+n_k=N \\ n_j\geq 1}} \frac{1}{(n_1\cdots n_k)^m}. \tag{5.3}$$

Theorem 5.1 (Exact sectors and an analytic certificate)**.** *The germ $\delta(q) = \sum_{N\geq 1} d_N^{(m)}(u) q^N$ has coefficients*

$$d_N^{(m)}(u) = (-1)^N \sum_{k=1}^{N} \frac{(-1)^{(m+1)k} C_{N,k}^{(m)}}{k!}\,\frac{\mathrm{d}^{k-1}}{\mathrm{d}z^{k-1}}\left(\mathrm{e}^{-Nz} Q_{m,u}(z)^{-k}\right)\Big|_{z=0}. \tag{5.4}$$

For a convergence certificate, choose $r > 0$ such that $P_m(u+z) - P_m(u)$ has exactly one zero in $|z| \leq r$, namely the simple zero at zero, and no boundary zero. Set

$$L = \min_{|z|=r} |P_m(u+z) - P_m(u)|.$$

If $\rho \mathrm{e}^r < 1$ and

$$\frac{\rho \mathrm{e}^r}{1-\rho \mathrm{e}^r} < L, \tag{5.5}$$

then δ is analytic for $|q| \leq \rho$, satisfies $|\delta| < r$, and, for $|q| < \rho$,

$$\left|\delta(q) - \sum_{n=1}^{N} d_n^{(m)}(u) q^n\right| \leq r\frac{(|q|/\rho)^{N+1}}{1-|q|/\rho}. \tag{5.6}$$

For each fixed $m \geq 2$, $r = 1/2$ and $\rho = 1/4$ satisfy the certificate for all sufficiently large u. Thus the physical inverse has a genuinely convergent exponential-sector representation at large positive core parameter.

Proof. The implicit derivative in δ at $(q,\delta)=(0,0)$ is $P'_m(u)>0$, so the analytic germ exists and is unique. Insert an auxiliary formal parameter λ into

$$z=\lambda\,\frac{(-1)^{m+1}\,\mathrm{Li}_m(-q\mathrm{e}^{-z})}{Q_{m,u}(z)}.$$

Lagrange inversion, first in λ, gives the kth derivative term in (5.4). The coefficient of q^N in the kth power of the polylogarithm is $(-1)^N C^{(m)}_{N,k}\mathrm{e}^{-Nz}$, and it is zero for $k>N$. Therefore setting $\lambda=1$ is legitimate coefficientwise in the q-adic ring and gives the finite formula. This is a specialization of the general contour reversion mechanism already developed in [10].

For $|z|\le r$, $|q|\le\rho$, the perturbation is analytic and has modulus at most $\rho\mathrm{e}^r/(1-\rho\mathrm{e}^r)$, since $m\ge 1$. Rouché's theorem and (5.5) give exactly one zero, counting multiplicity, of (5.1) in the disk. It is simple, hence depends holomorphically on q. Uniqueness glues the local solutions on the parameter disk. Cauchy's estimate gives $|d_n^{(m)}|\le r\rho^{-n}$ and summing a geometric tail proves (5.6).

For fixed r and $m\ge 2$,

$$P_m(u+z)-P_m(u)=P'_m(u)z\,(1+O(u^{-1}))$$

uniformly on $|z|\le r$, and $P'_m(u)\sim u^{m-1}/(m-1)!$. Comparison with this linear function certifies the single core zero and makes $L\to\infty$. The stated fixed r,ρ then work. At $q=\mathrm{e}^{-u}$ the root gives the inverse of F_m by (2.11); it is the real branch near u. Its local monotonicity also follows from $F'_m=F_{m-1}>0$ for $m\ge 2$ on the real axis. □

For $m=1$ the solution can be found without a certificate:

$$\delta(q)=\log(1-q)=-\sum_{N\ge 1}\frac{q^N}{N},\qquad |q|<1. \tag{5.7}$$

This checks both the signs and the normalization in (5.4). The general certificate is sufficient; it is not asserted to give an exact radius. The next sections replace it by a sharp result for $m=2$.

6 Dilogarithm sectors are positive Laurent polynomials

For $m=2$, the exact-core equation is

$$u\delta+\frac{\delta^2}{2}+\mathrm{Li}_2(-q\mathrm{e}^{-\delta})=0,\qquad \delta(0)=0. \tag{6.1}$$

Reflecting both variables gives a positive equation. Namely, let

$$v_u(r)=-\delta_u(-r). \tag{6.2}$$

Then

$$uv-\frac{v^2}{2}=\mathrm{Li}_2(r\mathrm{e}^v),\qquad v=\Phi_u(r,v):=\frac{v^2}{2u}+\frac{\mathrm{Li}_2(r\mathrm{e}^v)}{u}. \tag{6.3}$$

Theorem 6.1 (Positive finite formula)**.** *For every $u>0$, write $v_u(r)=\sum_{N\ge 1}a_N(u)r^N$. With $C_{N,k}=C^{(2)}_{N,k}$,*

$$a_N(u)=\sum_{k=1}^{N}\frac{C_{N,k}}{k}\sum_{j=0}^{k-1}\binom{k+j-1}{j}\frac{N^{k-1-j}}{(k-1-j)!\,2^j\,u^{k+j}}. \tag{6.4}$$

Every coefficient of every power $u^{-1}, \dots, u^{-(2N-1)}$ is a strictly positive rational number, and no other powers occur. The two extreme coefficients are

$$[u^{-1}]a_N(u) = \frac{1}{N^2}, \qquad [u^{-(2N-1)}]a_N(u) = \frac{\mathrm{Cat}_{N-1}}{2^{N-1}}, \quad \mathrm{Cat}_j = \frac{1}{j+1}\binom{2j}{j}. \tag{6.5}$$

Consequently $d_N^{(2)}(u) = (-1)^{N-1}a_N(u)$.

Proof. Here $Q_{2,u}(z) = u + z/2$. In the $(k-1)$st derivative of $\mathrm{e}^{-Nz}(u+z/2)^{-k}$ every Taylor coefficient contributing to that order has sign $(-1)^{k-1}$. Expanding the exponential and the negative binomial power in (5.4) therefore gives (6.4), with total sign $(-1)^{N-1}$.

The rational numbers $C_{N,k}$ are positive whenever $1 \le k \le N$. For any integer $1 \le p \le 2N-1$, one can choose $k \le N$ and $0 \le j < k$ with $p = k + j$, so every indicated power occurs positively. The power u^{-1} comes only from $k = 1, j = 0$, where $C_{N,1} = N^{-2}$. The highest power comes only from $k = N, j = N-1$, where $C_{N,N} = 1$. Its coefficient is $\binom{2N-2}{N-1}/(N2^{N-1})$, which is the Catalan expression. □

The first three coefficients are

$$\begin{aligned} a_1(u) &= \frac{1}{u}, \\ a_2(u) &= \frac{1}{4u} + \frac{1}{u^2} + \frac{1}{2u^3}, \\ a_3(u) &= \frac{1}{9u} + \frac{3}{4u^2} + \frac{7}{4u^3} + \frac{3}{2u^4} + \frac{1}{2u^5}. \end{aligned} \tag{6.6}$$

A separate exact recurrence, obtained by iterating the positive map Φ_u, verifies the finite formula independently of Lagrange inversion in the accompanying program. The positivity, rather than merely the existence of a coefficient formula, is the key to the global radius proof.

7 The exact radius and its unique dominant singularity

Theorem 7.1 (Sharp dilogarithm reversion radius)**.** *Fix $u > 0$. There is a unique $t = t_u \in (0,1)$ satisfying*

$$u^2 = 2\,\mathrm{Li}_2(t) + \log^2(1-t). \tag{7.1}$$

Set

$$w = -\log(1-t), \qquad v_c = u - w, \qquad R = R(u) = te^{w-u} = \mathrm{e}^{-u}\frac{t}{1-t}. \tag{7.2}$$

Then $0 < w < u$, $v_c > 0$, and the Taylor series of v_u has exact radius R. The point $r = R$ is its unique singularity on the circle $|r| = R$. In a slit neighborhood of R,

$$v_u(r) = v_c - \sqrt{2w(1-t)}\sqrt{1 - r/R} + O(1 - r/R), \tag{7.3}$$

with a convergent Puiseux expansion in $\sqrt{1-r/R}$. The coefficients satisfy

$$a_N(u) = \sqrt{\frac{w(1-t)}{2\pi}}\, R^{-N} N^{-3/2}(1 + O(N^{-1})). \tag{7.4}$$

All constants and error terms here are for fixed u. Equivalently, the unique dominant singularity of $\delta_u(q)$ is at $q = -R$.

Proof. The real characteristic point. The function $2\,\mathrm{Li}_2(t) + \mathrm{Li}_1(t)^2$ is strictly increasing from zero to infinity on $(0,1)$, since its derivative is positive and $\mathrm{Li}_1(t) = -\log(1-t) \to \infty$. Thus (7.1) has a unique solution. It implies $w^2 < u^2$ and $v_c > 0$. For $0 \le t \le t_u$ define

$$v(t) = u - \sqrt{u^2 - 2\,\mathrm{Li}_2(t)}, \qquad r(t) = t\mathrm{e}^{-v(t)}. \tag{7.5}$$

The radicand is positive on this whole interval. Differentiation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\log r(t) = \frac{1}{t}\left(1 - \frac{\mathrm{Li}_1(t)}{\sqrt{u^2 - 2\,\mathrm{Li}_2(t)}}\right). \tag{7.6}$$

It is positive for $0 < t < t_u$ and vanishes at t_u. Therefore (7.5) supplies a real analytic solution for $0 \le r < R$, tending to v_c at R.

No smaller convergence radius. Starting from $v_0 = 0$, iterate $v_{j+1} = \Phi_u(r, v_j)$ as a formal power series. Coefficients are nonnegative and increase coefficientwise. The coefficient of each fixed degree stabilizes, because $\partial_v \Phi_u(0,0) = 0$ makes the difference between two successive iterates gain at least one order in r. The limit is the unique formal solution. For fixed real $0 \le r < R$, the iterates are bounded above by the real solution in (7.5), since Φ_u preserves order on the relevant positive interval. Monotone convergence of the nonnegative coefficients shows that the full series converges for every such r, hence absolutely for $|r| < R$. Its sum is the real branch just constructed, by uniqueness near zero and analytic continuation. Moreover, monotone convergence as $r \uparrow R$ gives

$$\sum_{N \ge 1} a_N(u) R^N = v_c < \infty. \tag{7.7}$$

The singularity is a genuine square-root fold. At (R, v_c) the defining equation $v = \Phi_u(r, v)$ satisfies

$$\Phi_v = 1, \qquad R\Phi_r = \frac{w}{u}, \qquad \Phi_{vv} = \frac{1 + \mathrm{Li}_0(t)}{u} = \frac{1}{u(1-t)}. \tag{7.8}$$

Both nonzero derivatives are positive. The analytic implicit function theorem applied after solving for r as a function of v, followed by factoring its double zero at v_c, gives a convergent square-root inverse. In more detail, write $r = R - \alpha(v - v_c)^2 + O((v - v_c)^3)$ with $\alpha > 0$; the analytic square root of the nonvanishing factor reduces this to an ordinary analytic inverse in $\sqrt{R - r}$. Balancing the first nonzero terms in $v - \Phi_u(r, v) = 0$ gives the coefficient $\sqrt{2R\Phi_r/\Phi_{vv}} = \sqrt{2w(1-t)}$. The negative sign follows from $v(r) < v_c$ for $r < R$. This proves (7.3) and shows that the radius is not larger than R.

Uniqueness on the dominant circle. By (7.7), the series converges uniformly and absolutely on the closed disk. For $|r| = R$, $r \ne R$, strict triangle inequality gives $|v(r)| < v_c$, since every coefficient a_N is positive and the positive support includes all $N \ge 1$. Consequently

$$|r\mathrm{e}^{v(r)}| = R\mathrm{e}^{\Re v(r)} < R\mathrm{e}^{v_c} = t_u < 1.$$

The defining polylogarithm is analytic there, and

$$|\Phi_v(r, v(r))| \le \frac{|v(r)| + \mathrm{Li}_1(R\mathrm{e}^{\Re v(r)})}{u} < \frac{v_c + w}{u} = 1.$$

The implicit function theorem thus extends v analytically through every other point of the circle. A finite covering of its compact complement and the local square-root chart at R give a standard dented-disk (Δ-domain) continuation about the unique dominant singularity.

Coefficient asymptotics. The elementary singular term satisfies

$$[r^N](1 - r/R)^{1/2} = -\frac{R^{-N}}{2\sqrt{\pi}}\, N^{-3/2}(1 + O(N^{-1})).$$

The convergent Puiseux expansion and the preceding Δ-continuation permit termwise singularity transfer [6]. Analytic integer-power terms contribute no competing singular term; the next half-integer power has coefficient $O(R^{-N}N^{-5/2})$. This proves (7.4). The relation (6.2) transports the singularity to $q = -R$. □

Remark 7.2 (Why a real fold alone would not suffice). Solving the two real characteristic equations would provide a candidate radius, but would not exclude a closer complex singularity. The positive iteration and the strict boundary derivative estimate in the proof are what certify the *exact* radius and its unique dominant point.

8 The sharp physical threshold and an inverse identity

Theorem 8.1 (Necessary and sufficient physical convergence). *Let $y > \pi^2/6$ and u be given by (1.4). Then*

$$F_2^{-1}(y) = u + \sum_{N\geq 1} (-1)^{N-1} a_N(u) \mathrm{e}^{-Nu} \tag{8.1}$$

is an absolutely convergent identity if and only if

$$u \geq \frac{\pi}{\sqrt{6}}, \qquad \textit{equivalently} \qquad y \geq \frac{\pi^2}{4}. \tag{8.2}$$

At equality convergence is absolute despite evaluation on the Taylor circle. Below the threshold, the terms on the right do not tend to zero.

Proof. The physical parameter is $q = \mathrm{e}^{-u}$. By (7.2), $q \leq R$ is equivalent to $t/(1-t) \geq 1$, or $t \geq 1/2$. The characteristic function in (7.1) is strictly increasing. The classical reflection identity gives

$$\mathrm{Li}_2(1/2) = \frac{\pi^2}{12} - \frac{\log^2 2}{2}, \qquad 2\,\mathrm{Li}_2(1/2) + \log^2 2 = \frac{\pi^2}{6}.$$

Thus $q \leq R$ is exactly (8.2). For $q < R$, the series is the analytic germ from (6.1). For every real $q > 0$, that equation has a unique positive solution: its left side is negative at $\delta = 0$, tends to infinity, and has strictly positive δ derivative

$$u + \delta + \log(1 + q\mathrm{e}^{-\delta}) > 0.$$

Consequently the analytic germ is this real root wherever its Taylor series converges, and (2.11) identifies the root with the required inverse at $q = \mathrm{e}^{-u}$.

At $q = R$, (7.7) or (7.4) gives absolute convergence. Continuity extends the same defining equation and hence the same inverse identity to the endpoint. If $q > R$, (7.4) gives $a_N q^N \sim C(u)(q/R)^N N^{-3/2}$, with $C(u) > 0$; terms do not tend to zero. Finally $y = u^2/2 + \pi^2/6$ converts the threshold in u to that in y. □

The derivative $F_2'(x) = \log(1 + \mathrm{e}^x) > 0$ and the limits at $\pm\infty$ show that F_2 maps $\mathbb{R}$ bijectively onto $(0, \infty)$. Therefore the failure below (8.2) is solely a failure of this expansion, not of the real inverse.

Corollary 8.2 (An exact self-referential radius identity). *For every $u > 0$,*

$$R(u) = \exp(F_2^{-1}(u^2/2) - u). \tag{8.3}$$

Also $0 < R(u) < 1$, $R(u) \sim u^2/2$ as $u \downarrow 0$, and

$$\log R(u) = -\frac{\pi^2}{6u} + O(u^{-3}) \qquad (u \to \infty). \tag{8.4}$$

Proof. Let $a = t/(1-t) = \mathrm{e}^w - 1$. The classical Landen identity, obtained also by differentiating both sides and matching at $t = 0$, is

$$-\operatorname{Li}_2\!\left(-\frac{t}{1-t}\right) = \operatorname{Li}_2(t) + \frac{1}{2}\log^2(1-t).$$

The characteristic equation therefore gives $F_2(\log a) = u^2/2$. Since $R = \mathrm{e}^{-u}a$, (8.3) follows. It is an exact functional relation, not an independent shortcut for computing an inverse one has not yet evaluated.

The inequalities $t < 1$ and $w < u$ give $R = t\mathrm{e}^{w-u} < 1$. In w coordinates the characteristic equation is

$$u^2 = w^2 + 2\operatorname{Li}_2(1 - \mathrm{e}^{-w}). \tag{8.5}$$

Its right side has derivative $2w/(1-\mathrm{e}^{-w})$, with limiting value 2 at zero. Hence $w \sim u^2/2$ and $R = \mathrm{e}^{-u}(\mathrm{e}^w - 1) \sim u^2/2$. For large u, $w \to \infty$, and the reflection identity gives $\operatorname{Li}_2(1-\mathrm{e}^{-w}) = \pi^2/6 + O(w\mathrm{e}^{-w})$. Thus $w = u - \pi^2/(6u) + O(u^{-3}) + O(\mathrm{e}^{-w})$. Using $\log R = w - u + \log(1-\mathrm{e}^{-w})$ proves (8.4). □

9 A quadratic polylogarithm-feedback extension

The sharp-radius argument depends more on positivity than on the special functional identities of the dilogarithm. Those identities are what turn the general radius criterion into the especially simple threshold $\pi^2/4$.

Theorem 9.1 (Interior folds for quadratic polylogarithm feedback)**.** *Let $s > 0$, $u > 0$, and let v be the germ with $v(0) = 0$ satisfying*

$$uv - \frac{v^2}{2} = \operatorname{Li}_s(r\mathrm{e}^v). \tag{9.1}$$

Suppose there is $t \in (0,1)$ with

$$u^2 = 2\operatorname{Li}_s(t) + \operatorname{Li}_{s-1}(t)^2. \tag{9.2}$$

This condition holds for every $u > 0$ when $0 < s \le 2$. For $s > 2$ it holds exactly when

$$u < U_c(s) := \sqrt{2\zeta(s) + \zeta(s-1)^2}. \tag{9.3}$$

The solution t is unique. Put

$$w = \operatorname{Li}_{s-1}(t), \quad v_c = u - w, \quad R = t\mathrm{e}^{w-u}. \tag{9.4}$$

Then the Taylor series of v has positive coefficients, exact radius R, and unique dominant singularity R. Its coefficient asymptotic is

$$[r^N]v(r) \sim \sqrt{\frac{w}{2\pi(1 + \operatorname{Li}_{s-2}(t))}}\, R^{-N} N^{-3/2}. \tag{9.5}$$

The positive finite formula (6.4) remains valid after replacing $C_{N,k}$ by $[z^N]\operatorname{Li}_s(z)^k$. Its coefficients in u^{-1} are then positive real numbers; the extreme coefficients are N^{-s} and $\operatorname{Cat}_{N-1}/2^{N-1}$.

Proof. For real s, each coefficient n^{-s} is positive. All derivatives with respect to t of a polylogarithm in the open interval $(0,1)$ are therefore positive. In particular, $2\operatorname{Li}_s(t)+\operatorname{Li}_{s-1}(t)^2$ is strictly increasing. Its limit at $t=1$ is infinite if $s\le 2$, and is $U_c(s)^2$ if $s>2$. This proves the existence and uniqueness assertions.

Use the same parameterization as (7.5), with Li_s in place of Li_2. Its logarithmic r derivative is $t^{-1}(1-\operatorname{Li}_{s-1}(t)/\sqrt{u^2-2\operatorname{Li}_s(t)})$, positive before the characteristic point. Iterating $\Phi(r,v)=v^2/(2u)+\operatorname{Li}_s(re^v)/u$ gives nonnegative formal coefficients and bounds their sum up to R exactly as in the proof of Theorem 7.1. The Lagrange calculation in Theorem 6.1 uses only the quadratic core, so its positive finite formula applies unchanged.

At the fold,

$$R\Phi_r = w/u, \qquad \Phi_{vv} = (1+\operatorname{Li}_{s-2}(t))/u > 0.$$

Hence the square-root coefficient is $\sqrt{2w/(1+\operatorname{Li}_{s-2}(t))}$. Every coefficient is positive. On another point of the circle, $|v|<v_c$, $|re^v|<t$, and $|\Phi_v|\le(|v|+\operatorname{Li}_{s-1}(|re^v|))/u<1$. This proves unique dominance and the same Δ continuation as before. The square-root transfer gives (9.5). □

For $0<s\le 2$, the boundary between convergence and divergence on the curve $r=-e^{-u}$ can likewise be stated exactly. The unique $t_*\in(0,1)$ solving

$$\log t_* + \operatorname{Li}_{s-1}(t_*) = 0 \tag{9.6}$$

gives $u_*^2 = 2\operatorname{Li}_s(t_*)+\operatorname{Li}_{s-1}(t_*)^2$. Convergence holds precisely for $u\ge u_*$, including absolute convergence at the boundary. This follows from $e^{-u}/R = e^{-w}/t$ and monotonicity of both sides of the characteristic parametrization. For $s=2$, $t_*=1/2$. For other values of s, (9.1) is a quadratic feedback model, *not* the exact-core inversion equation of F_s; that identification is special to the dilogarithm case.

10 An audit safeguard: arithmetic averaging and inversion

The repository source [10] already explicitly studies the nonlinear real-averaging defect. The calculation below is an illustrative restatement and integration safeguard, not a newly discovered phenomenon. The exact identity (2.10) uses an arithmetic average of lateral realizations. It must not be propagated through nonlinear inversion as though averaging were a homomorphism. The following elementary calculation isolates the obstruction.

Proposition 10.1 (First nonlinear averaging defect)**.** *Let F be analytic and locally invertible with inverse g, let T be analytic near the relevant inverse point, and let $G_\pm(y;\lambda)$ be the coherently chosen inverses of $F\pm \mathrm{i}\lambda T$ for sufficiently small λ. Then*

$$\frac{G_+ + G_-}{2} = g - \frac{\lambda^2}{2}\frac{\mathrm{d}}{\mathrm{d}y}\big(g'(y)T(g(y))^2\big) + O(\lambda^4), \tag{10.1}$$

locally uniformly. The inverse of the arithmetic average of the forward functions is exactly g, so these operations need not commute.

Proof. The analytic perturbation inverse has the convergent local expansion

$$G_\pm = g + \sum_{n\ge 1} \frac{(\mp\mathrm{i}\lambda)^n}{n!}\frac{\mathrm{d}^{n-1}}{\mathrm{d}y^{n-1}}\big(g'(y)T(g(y))^n\big).$$

It follows either by a local contour logarithm, as in [10], or by Lagrange inversion in the forward coordinate. Averaging removes the odd terms; the $n=2$ coefficient is the displayed one. Holomorphy in λ gives the uniform fourth-order remainder. □

For $F(x) = x$ and $T(x) = \mathrm{e}^{-x}$, the defect is $+\lambda^2\mathrm{e}^{-2y} + O(\lambda^4)$. This gives an explicit counterexample, not just a warning based on formal terminology. It is relevant when the forward polylogarithm happens to equal an arithmetic lateral average, as at a half-integer order in (2.10).

This proposition is *not* a counterexample to appropriately defined Écalle median summation or to a summation morphism whose hypotheses hold. Arithmetic averaging and such nonlinear-compatible summation procedures are different constructions. Nor was an erroneous averaging claim found in the inspected accepted repository theorem. The proposition is a preventive integration note, not an allegation about an unseen text.

11 Verification, limitations, and proposed integration

11.1 Exact replay and numerical diagnostics

The distributed program `code/verify.py` uses Python rational arithmetic for the inverse coefficients and SymPy for finite symbolic checks. At the recorded run it passed **118 exact assertions**: positivity and complete Laurent support through $N = 10$, both extreme coefficients at each of those orders, thirty coefficient comparisons against an independent fixed-point recurrence at $u = 1, 2, 3$, and finite Taylor/residue checks of six Gamma kernels. These assertions are finite regressions, not a formal proof of a universally quantified theorem.

The arbitrary-precision run used `mpmath` at 65 decimal digits. Three complex-ray reconstructions of the first order derivative were compared with differentiation under the defining Fermi integral, twelve higher-jet coefficient comparisons were performed, and six periodic kernels were checked in both digamma and convergent Hurwitz-series coordinates. Their Fourier tails were separately checked against the exponential Dirichlet series. The recorded absolute errors for these comparisons were below 10^{-53}; this is floating-point agreement, not an interval certificate.

The radius equation was solved monotonically in its w coordinate. Table 1 displays selected results, and the full-precision records are in `data/radii.csv`. In particular, a ratio larger than one puts the physical point outside the convergence disk.

Table 1: Exact-radius diagnostics, rounded from 65-digit calculations.

u	w	$R(u)$	$\mathrm{e}^{-u}/R(u)$
0.7	0.231283506580	0.129219674046	3.842954313697
1	0.447460986781	0.207607344120	1.771996278510
$\pi/\sqrt{6}$	$\log 2$	0.277329255639	1
2	1.418886391532	0.423939922809	0.319232221254
4	3.600624855684	0.652423391327	0.028073240678
8	7.792136744823	0.811982646785	0.000413140144

Table 2 checks the fixed-u asymptotic constant by positive convolution and the finite formula; these particular larger-N calculations use binary64, not 65-digit arithmetic. Their slower approach at $u = 4$ is consistent with, but does not prove, the lack of uniformity as $u \to \infty$.

Direct inverse comparisons at $u = 2$ and $u = 4$ used up to 24 exact coefficient polynomials. At $N = 24$, the absolute inverse errors were about $5.74 \cdot 10^{-16}$ and $1.72 \cdot 10^{-42}$ respectively. The associated forward residuals were also checked. These are evaluations of a proved convergent expansion; they are not the source of the convergence theorem.

Table 2: The normalized coefficient $a_N(u)R(u)^N N^{3/2}/\sqrt{w(1-t)/(2\pi)}$, which tends to one.

N	$u = 2$	$u = 4$
16	0.994010739	1.119387718
32	0.997036572	1.078796636
64	0.998528610	1.049118033
128	0.999267319	1.029009229
256	0.999634478	1.016318713

11.2 A numerical differentiation trap detected during the audit

A preliminary check used a tiny-step numerical order derivative of $-\operatorname{Li}_s(-\mathrm{e}^{1.5})$ at $s = 1$. With the installed `mpmath 1.3.0`, that route did not retain the requested accuracy. At 50-digit working precision, one observed value was

$$0.90405058162286877632141113281 25,$$

whereas both the independent defining-integral derivative and the Borel reconstruction gave

$$0.90405057957314917027110174646868381462250001519 1\ldots.$$

The discrepancy is approximately $2.05 \cdot 10^{-9}$; merely quoting the working precision would conceal it. This is an observed behavior of that specific evaluation route and version, not a general theorem about the library or a diagnosis of its internal cause. The final verification program does *not* use that route as its reference. It evaluates

$$J_m(x) = \frac{1}{\Gamma(m)} \int_0^\infty \frac{t^{m-1}(\log t - \psi(m))}{1 + \mathrm{e}^{t-x}}\, \mathrm{d}t, \tag{11.1}$$

which follows directly from (2.1). The package includes a small reproducer and a version-specific note so the failed preliminary check is not silently lost.

11.3 Source audit and integration boundary

The audit inspected selected portions of the project READMEs, the initial portion of the polylogarithm–polygamma bridge article, the nonlinear Stokes-transport title/abstract, proof, and countable-family/flow discussion, and the critical-Hahn package README. It did not inspect every chapter of the large canonical volumes or every pending incoming article. The last observed main commit was `cbfbc3163e0d038a381808c6db8d39ea12dc0db7`; individual file blob identifiers and reading ranges are recorded in `SOURCES.md`. Some initial reads preceded that commit observation and were made on `main`; the audit is therefore not falsely presented as an atomic full-tree checkout.

No new mathematical error was established in the accepted repository statements actually inspected. The proposed corrections are narrowly scoped safeguards for this contribution: retain exponential completion at integer resonance; retain order derivatives of vanishing coefficients; fix the lateral sign convention; preserve the zero Fourier mode; do not replace a sufficient certificate by an exact-radius claim; and do not commute arithmetic averaging with inversion. Existing source warnings about evidence versus proof are respected, not rebranded as discoveries.

Recommended placement is a new self-contained research package under `Analysis/Transseries/docs/series-and-transseries/`, with a backlink from the polylogarithm reports. The generic inverse engine should remain attributed to its existing source. The

sharper, model-specific radius and periodic-kernel statements can be considered separately for promotion to the canonical volumes after independent proof review. The package changes no repository files remotely.

12 Further research questions and concrete next steps

1. The boundary regime of quadratic polylogarithm feedback. Theorem 9.1 stops at an interior fold. For $s > 2$ and $u \geq U_c(s)$, the polylogarithm boundary at $t = 1$ competes with or replaces the fold. A useful precise target, when $u > U_c(s)$, is the candidate law

$$[r^N]v(r) \sim \frac{w_0^{s-1}}{(w_0 - \zeta(s-1))^s} R_b^{-N} N^{-s}, \qquad w_0 = \sqrt{u^2 - 2\zeta(s)}, \quad R_b = \mathrm{e}^{-(u-w_0)}.$$

This expression follows from formal transport of the leading polylogarithm singular term, but is *not proved here*. A proof must establish the relevant boundary continuation, handle integer-order logarithms, and exclude competing singularities. At $u = U_c(s)$, one expects different exponents for $2 < s < 3$, a logarithmic transition at $s = 3$, and a finite-curvature regime above it. The related linear-core critical-Hahn work in [11] should be used as a comparison source rather than ignored.

2. Joint large-sector and large-core asymptotics. The constant in (7.4) tends rapidly to zero as $u \to \infty$, while $R(u) \to 1$. Determine a uniform law for $N, u \to \infty$, including the transition from the fixed-N behavior $a_N(u) \sim 1/(N^2 u)$ to the fixed-u square-root law. The parameter $1 - t_u = \mathrm{e}^{-w_u}$ measures the approaching polylogarithm boundary and suggests that the product of this quantity with the sector index deserves particular attention; the correct full scaling should be derived, not assumed from the separate limits.

3. Exact radii for higher integer-order cores. For $m \geq 3$, (5.4) is finite and explicit, but $Q_{m,u}$ is no longer linear. Classify when an analogous sign pattern survives, locate the nearest complex critical value, and determine whether any further physical thresholds reduce to known special polylogarithm values. The sufficient certificate in Theorem 5.1 provides a safe starting region but does not answer those questions.

4. A meromorphic–logarithmic dictionary for higher jets. Theorem 3.2 gives every coefficient, whereas Theorem 3.1 gives a particularly simple global Borel kernel only for the first jet. Derive comparable global representations for $r \geq 2$, with the logarithmic singular germs at every integer made explicit. The differentiated lattice formula is available, but an arithmetic normal form should keep the harmonic sums and branch constants separate and reveal which finite Gamma/polygamma coordinates suffice.

5. Resonance detuning coupled to nonlinear inversion. Equation (3.21) resolves the moving order at the coefficient level. Construct a uniform inverse chart when the order approaches an integer while the core grows, and determine the scale at which the restored divergent tail competes with the pre-existing exponential inverse sectors. Coefficientwise convergence alone does not justify commuting these limits with Borel integration and reversion.

6. Arithmetic cancellation of the first Borel actions. Periodic weights may cancel the first several residues exactly. Determine how such cancellations change the first visible inverse sectors, the nearest surviving Borel singularities, and optimal truncation. The finite DFT in Theorem 4.1 makes this a concrete algebraic design problem. A nonzero mean must remain in

the calculation; cancellation of some residue classes does not imply disappearance of the zero mode.

7. Nonlinear-compatible real summation. Starting with the explicit jump (2.9), construct and compare real analytic completions whose inverse transport is compatible with a chosen summation framework. Proposition 10.1 supplies a low-order obstruction test. The goal is an actual compatible construction, not a relabeling of arithmetic averaging as a median operation.

8. Formalization and certified computation. A practical first formalization target is the positive coefficient formula and its two extreme coefficients over $\mathbb{Q}[u^{-1}]$, followed by the finite-Fourier residue dictionary. The analytic radius theorem additionally needs a positive implicit-series argument, a complex continuation API, and a square-root transfer theorem. Separately, replace the current floating-point Borel diagnostics by interval enclosures on tilted rays with rigorous tail bounds. These tasks are proposed work; the present package contains neither a Lean proof nor an interval certificate.

A Reproduction guide and result map

All manuscript formulas are contained in this one `article.tex`; no external image or font file is required. The retained tables are rounded from the accompanying CSV files. The following commands reproduce the verification and the PDF in a standard Python/TeX environment:

```
python -m pip install -r requirements.txt
python code/verify.py --output-dir replay --max-n 256
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

Use a new output directory for a replay rather than overwriting historical evidence. The optional `-skip-borel` switch is explicitly recorded as a skipped check; it was *not* used in the final retained run.

Result	Status in this article	Dependency and scope
Theorem 2.2	Classical completion, explicit realization	Defining integral, binomial series, normal convergence; no claim that the complete Sommerfeld formula is new.
Theorems 3.1 and 3.2	Derived jet identities	Gamma reflection, analytic order differentiation, explicit endpoint terms; conventional proofs.
Theorem 4.1	Derived arithmetic kernel dictionary	Subtracted digamma resolvent and finite DFT; zero mean not assumed.
Theorem 5.1	Model specialization of existing tools	Lagrange inversion and Rouché certification; sufficient radius only.
Theorems 6.1–8.1	Main proposed contribution	All-sector positivity, exact radius, unique dominance, sharp endpoint; not established merely by the numerical checks.
Theorem 9.1	Derived interior-fold extension	All $0 < s \leq 2$, or $s > 2$ below the stated boundary; no boundary theorem claimed at or above $U_c(s)$.

Result	Status in this article	Dependency and scope
Proposition 10.1	Exact audit safeguard	Elementary perturbative inverse calculation; not a refutation of median summation theory.

References

[1] NIST Digital Library of Mathematical Functions, *§25.12, Polylogarithms*, especially the integral representation, Jonquière expansion, dilogarithm reflection and Landen identities. https://dlmf.nist.gov/25.12. Accessed 11 October 2026.

[2] NIST Digital Library of Mathematical Functions, *§5.5, Functional Relations*, Gamma reflection. https://dlmf.nist.gov/5.5. Accessed 11 October 2026.

[3] NIST Digital Library of Mathematical Functions, *§5.11, Asymptotic Expansions*, especially the digamma expansion and Gamma-ratio formulas. https://dlmf.nist.gov/5.11. Accessed 11 October 2026.

[4] T. M. Garoni, N. E. Frankel, and M. L. Glasser, Complete asymptotic expansions of the Fermi–Dirac integrals, *Journal of Mathematical Physics* **42** (2001), 1860–1868. https://doi.org/10.1063/1.1350634. The present source audit inspected the institutional publication record and abstract, not the complete article.

[5] A. Gil, J. Segura, and N. M. Temme, *Complete asymptotic expansions for the relativistic Fermi–Dirac integral*, arXiv:2108.11210v2, Section 3 and Lemma 3.1. https://arxiv.org/html/2108.11210v2.

[6] P. Flajolet and A. Odlyzko, Singularity analysis of generating functions, *SIAM Journal on Discrete Mathematics* **3** (1990), 216–240. https://doi.org/10.1137/0403019.

[7] ProveIt Contributors, *Polylogarithms: project README and claim-status overview*, `Analysis/Polylogarithms/README.md`, in the ProveIt repository. https://github.com/VladimirReshetnikov/ProveIt. Reading scope and observed revision are recorded in `SOURCES.md`.

[8] V. Reshetnikov, *The Polylogarithm–Polygamma Bridge: An Exact Finite Fourier Transform, Clausen-Type Laws at Rational Arguments, and the Layers Below the Digamma*, 11 June 2026, historical source retained at `Analysis/Polylogarithms/docs/articles/polylog-polygamma-bridge.tex`. The current project ledger, not the historical abstract alone, governs accepted claim status.

[9] ProveIt Contributors, *Series and transseries: canonical volume and research-package index*, `Analysis/Transseries/docs/series-and-transseries/README.md`, ProveIt repository, inspected 11 October 2026.

[10] ProveIt research package, *Nonlinear Stokes Transport under Logarithmic Inversion*, `Analysis/Transseries/docs/series-and-transseries/Nonlinear_Stokes_Transport_Logarithmic_Inversion/article.tex`. Inspected source lines 1–110 and 180–370; blob `1e8b24325554d1f1c94c29f21b794ba62feff755`. This is the source for the general contour inverse-transport architecture, its countable-family qualification, and the distinction between a certificate and a formal formula.

[11] ProveIt research package, *Beyond Finite-Action Folds: Critical Hahn Transseries, Stable Sector Asymptotics, and Sharp Action Budgets*, README at `Analysis/Transseries/docs/series-and-transseries/Critical_Hahn_Transseries_Beyond_Finite_Action_Folds/README.md`. Inspected at commit `cbfbc3163e0d038a381808c6db8d39ea12dc0db7`; README blob `7d138677987dd1cc6dbfdf5df1aa564bfd08d5dd`. Its linear-core model and editorial cross-references are comparison material, not analytic results audited in full here.